

Task 3

Мікросервиси з використанням Hazelcast Distributed Map

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[Github repo](#)

0. Setup

Система побудована на основі опису завдання й переймає більшість логіки з [Task 1](#). Для імплементації бази даних використовувався PostgreSQL. Приклад повного застосунку, запущеного в Docker:

```
PS C:\Users\Admin\Desktop\APZ\assignment_3> docker compose ps
>>

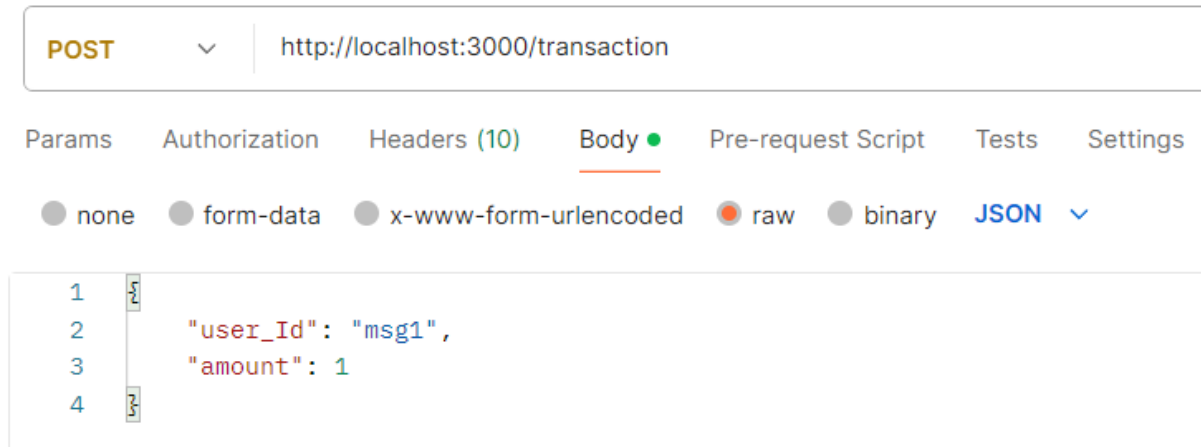
```

NAME	PORTS	IMAGE	COMMAND	SERVICE	CREATED	STATUS	
assignment_3-counter-service-1		assignment_3-counter-service	"docker-entrypoint.s..."	counter-service	About an hour ago	Up About an hour	0.0.0.0:3002->3002/tcp, [:::3002->3002/tcp]
assignment_3-facade-service-1		assignment_3-facade-service	"docker-entrypoint.s..."	facade-service	About an hour ago	Up About an hour	0.0.0.0:3000->3000/tcp, [:::3000->3000/tcp]
assignment_3-hazelcast-1-1		hazelcast/hazelcast:5.6.0	"hz start"	hazelcast-1	About an hour ago	Up About an hour	0.0.0.0:5701->5701/tcp, [:::5701->5701/tcp]
assignment_3-hazelcast-2-1		hazelcast/hazelcast:5.6.0	"hz start"	hazelcast-2	About an hour ago	Up About an hour	0.0.0.0:5702->5701/tcp, [:::5702->5701/tcp]
assignment_3-hazelcast-3-1		hazelcast/hazelcast:5.6.0	"hz start"	hazelcast-3	About an hour ago	Up About an hour	0.0.0.0:5703->5701/tcp, [:::5703->5701/tcp]
assignment_3-logging-service-1-1		assignment_3-logging-service-1	"docker-entrypoint.s..."	logging-service-1	About an hour ago	Up About an hour	0.0.0.0:3001->3001/tcp, [:::3001->3001/tcp]
assignment_3-logging-service-2-1		assignment_3-logging-service-2	"docker-entrypoint.s..."	logging-service-2	About an hour ago	Up 12 seconds	0.0.0.0:3003->3001/tcp, [:::3003->3001/tcp]
assignment_3-logging-service-3-1		assignment_3-logging-service-3	"docker-entrypoint.s..."	logging-service-3	About an hour ago	Up About an hour	0.0.0.0:3004->3001/tcp, [:::3004->3001/tcp]
assignment_3-postgres-1		postgres:16-alpine	"docker-entrypoint.s..."	postgres	About an hour ago	Up About an hour (healthy)	0.0.0.0:5432->5432/tcp, [:::5432->5432/tcp]

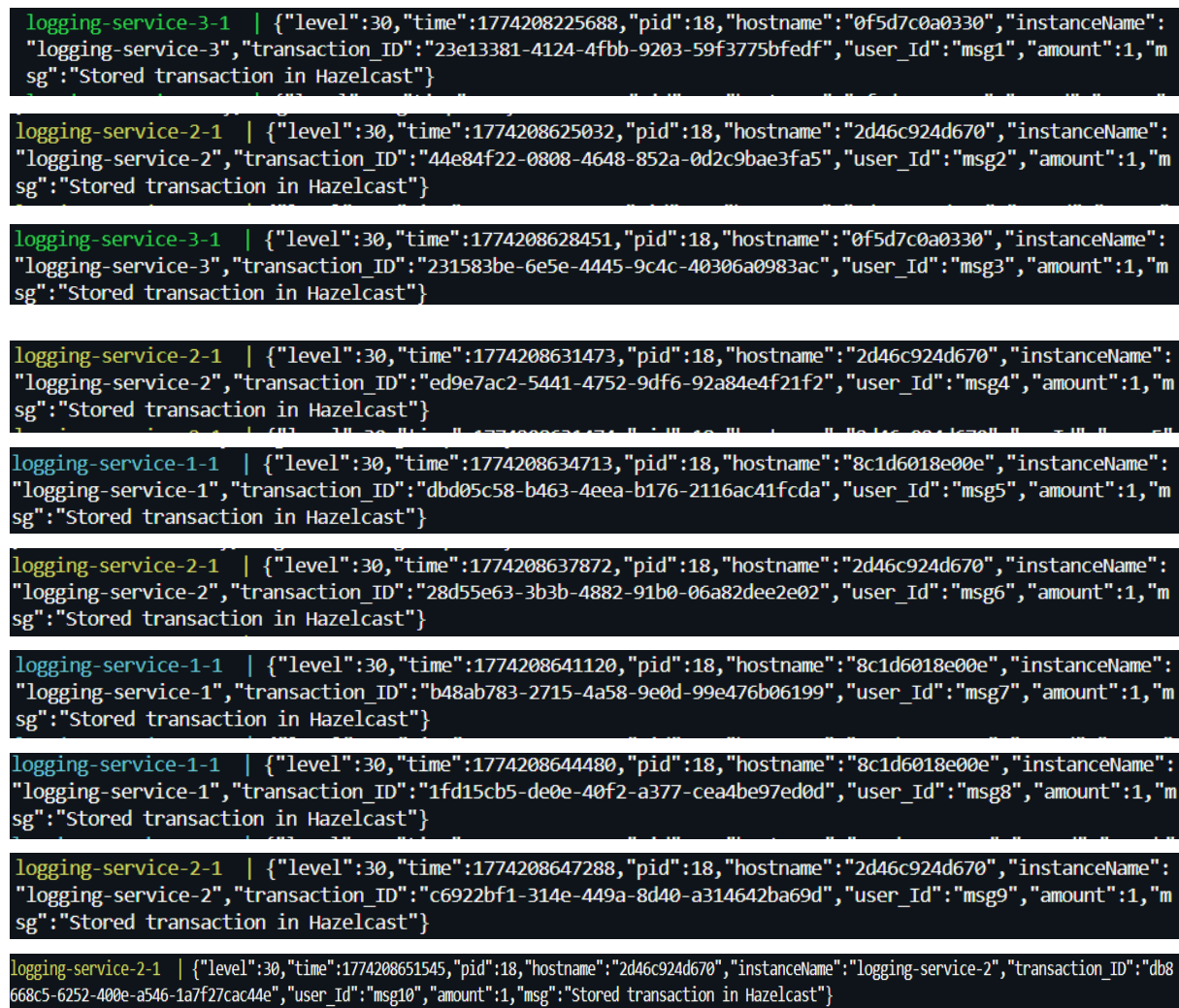
```
PS C:\Users\Admin\Desktop\APZ\assignment_3>
```

1. Базові HTTP запити:

Запити надсилалися через Postman. Приклад першого POST запиту (інші ідентичні):



Вігновігні логу в logging-service:



GET /accounts:

GET http://localhost:3000/accounts Send

Params Authorization Headers (10) Body ● Pre-request Script Tests Settings Cookies

Body Cookies Headers (5) Test Results Status: 200 OK Time: 119 ms Size: 277 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "accounts": [
3     {
4       "msg1": 1,
5       "msg10": 1,
6       "msg2": 1,
7       "msg3": 1,
8       "msg4": 1,
9       "msg5": 1,
10      "msg6": 1,
11      "msg7": 1,
12      "msg8": 1,
13      "msg9": 1
14    }
15  ]
16 }
```

Приклад GET /user/<user_Id> (інші ідентичні):

GET http://localhost:3000/user/msg1 Send

Params Authorization Headers (10) Body ● Pre-request Script Tests Settings Cookies

Body Cookies Headers (5) Test Results Status: 200 OK Time: 405 ms Size: 327 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "balance": 1,
3   "logs": [
4     {
5       "transaction_ID": "f068d46d-68b0-4efd-ae7-df09e4df3d7f",
6       "timestamp": "2026-03-22T19:43:35.855Z",
7       "msg": {
8         "user_Id": "msg1",
9         "amount": 1
10      }
11    }
12  ]
13 }
```

2. Перевірка відмовостійкості:

- Вимкнення logging-service-2 - результати POST/GET запитів та логи (система перезапущена після Пункту 1, тож попередні транзакції відсутні):

Приклад POST запиту (інші аналогічні):

POST http://localhost:3000/transaction

Params Authorization Headers (10) Body Pre-request Script Tests Settings Cookies Beautify

none form-data x-www-form-urlencoded raw binary JSON

```
1 {
2   "user_id": "test-1-1",
3   "amount": 10
4 }
```

Body Cookies Headers (5) Test Results Status: 200 OK Time: 63 ms Size: 241 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "transaction_ID": "75b872e1-749a-4e65-96e3-743ebc2d33eb",
3   "balance": 10
4 }
```

Логи logging-service:

```
logging-service-3-1 | {"level":30,"time":1774210377694,"pid":18,"hostname":"0f5d7c0a0330","instanceName":"logging-service-3","transaction_ID":"75b872e1-749a-4e65-96e3-743ebc2d33eb","user_id":"test-1-1","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-3-1 | {"level":30,"time":1774210408319,"pid":18,"hostname":"0f5d7c0a0330","instanceName":"logging-service-3","transaction_ID":"10ab6c66-0e16-430b-980e-affc3b59b001","user_id":"test-1-2","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-3-1 | {"level":30,"time":1774210456215,"pid":18,"hostname":"0f5d7c0a0330","instanceName":"logging-service-3","transaction_ID":"e12fb190-63eb-4546-a53f-3d50ed83d4b8","user_id":"test-1-3","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-1-1 | {"level":30,"time":1774210460415,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"a9986533-6858-4799-be64-e220ab4054b","user_id":"test-1-4","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-3-1 | {"level":30,"time":1774210464770,"pid":18,"hostname":"0f5d7c0a0330","instanceName":"logging-service-3","transaction_ID":"99e7c7a3-a29e-4573-99e9-08fc2b18cc5c","user_id":"test-1-5","amount":10,"msg":"Stored transaction in Hazelcast"}
```

GET /accounts:

HTTP http://localhost:3000/accounts Save

GET http://localhost:3000/accounts Send Cookies

Params Authorization Headers (10) Body Pre-request Script Tests Settings

Body Cookies Headers (5) Test Results Status: 200 OK Time: 56 ms Size: 255 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "accounts": {
3     "test-1-1": 10,
4     "test-1-2": 10,
5     "test-1-3": 10,
6     "test-1-4": 10,
7     "test-1-5": 10
8   }
9 }
```

Приклад GET /user/<user_Id> (інші ідентичні):

GET http://localhost:3000/user/test-1-1

Status: 200 OK Time: 2.33 s Size: 333 B

```
{
  "balance": 10,
  "logs": [
    {
      "transaction_ID": "75b872e1-749a-4e65-96e3-743ebc2d33eb",
      "timestamp": "2026-03-22T20:12:57.686Z",
      "msg": {
        "user_Id": "test-1-1",
        "amount": 10
      }
    }
  ]
}
```

Помилка при спробі доступитися до logging-service-2 (test-1-3, згодом успішно залогований у logging-service-3):

```
facade-service-1 | {"level":40,"time":1774210456204,"pid":18,"hostname":"c4527c291594","err":{"type":"TypeError","message":"fetch failed: getaddrinfo ENOTFOUND logging-service-2","stack":"TypeError: fetch failed\n at node:internal/deps/undici/undici:14902:13\n at process.processTicksAndRejections (node:internal/process/task_queues:95:5)\n at async fetchLoggingWithFailover (file:///app/dist/index.js:27:25)\n at async file:///app/dist/index.js:62:34\n at async Promise.all (index 0)\n at async Object.<anonymous> (file:///app/dist/index.js:87:33)\ncaused by: Error: getaddrinfo ENOTFOUND logging-service-2\n at GetAddrInfoReqWrap.onlookupall [as oncomplete] (node:dns:122:26)","replica":"http://logging-service-2:3001","path":"/transaction","msg":"Logging replica unavailable"}}
```

- Вимкнення logging-service-2 та logging-service-3 - результати POST/GET запитів та логи (система перезапущена після Пункту 1, мож попередні транзакції відсутні):

Приклад POST запиту (інші аналогічні):

POST http://localhost:3000/transaction

Status: 200 OK Time: 1860 ms Size: 241 B

```
{
  "transaction_ID": "80bde2d0-7619-4c93-88af-031df0d84329",
  "balance": 10
}
```

Λογυ logging-service:

```
logging-service-1-1 | {"level":30,"time":1774211946186,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"80bde2d0-7619-4c93-88af-031df0d84329","user_Id":"test-2-1","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-1-1 | {"level":30,"time":1774212096081,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"f1f6d157-b7b2-48c1-a9a6-22e1fb073c58","user_Id":"test-2-2","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-1-1 | {"level":30,"time":1774212099235,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"05461a8e-a576-406d-9ef7-d4b097b2ff02","user_Id":"test-2-3","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-1-1 | {"level":30,"time":1774212103002,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"9b146a32-fef9-4835-b2e0-14ced7a09807","user_Id":"test-2-4","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-1-1 | {"level":30,"time":1774212108727,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"60314c46-37f5-414f-acda-4793b286dd76","user_Id":"test-2-5","amount":10,"msg":"Stored transaction in Hazelcast"}
```

GET /accounts:

GET http://localhost:3000/accounts

Params Authorization Headers (10) Body Pre-request Script Tests Settings Cookies

Body Cookies Headers (5) Test Results Status: 200 OK Time: 58 ms Size: 326 B Save Response

Pretty Raw Preview Visualize JSON

```
1  {
2    "accounts": {
3      "test-1-1": 10,
4      "test-1-2": 10,
5      "test-1-3": 10,
6      "test-1-4": 10,
7      "test-1-5": 10,
8      "test-2-1": 10,
9      "test-2-2": 10,
10     "test-2-3": 10,
11     "test-2-4": 10,
12     "test-2-5": 10
13   }
14 }
```

Παράδειγμα GET /user/<user_Id> (ίδια ή διαφορετικά):

GET http://localhost:3000/user/test-2-2

Params Authorization Headers (10) Body Pre-request Script Tests Settings Cookies

Body Cookies Headers (5) Test Results Status: 200 OK Time: 3.76 s Size: 333 B Save Response

Pretty Raw Preview Visualize JSON

```
1  {
2    "balance": 10,
3    "logs": [
4      {
5        "transaction_ID": "f1f6d157-b7b2-48c1-a9a6-22e1fb073c58",
6        "timestamp": "2026-03-22T20:41:32.326Z",
7        "msg": {
8          "user_Id": "test-2-2",
9          "amount": 10
10        }
11      }
12    ]
13 }
```

- Вимкнення assignment_3-hazelcast-2:

```
logging-service-3-1 | [DefaultLogger] WARN at Connection: Connection{alive=false, connectionId=1, remoteAddress=hazelcast-2:5701} closed. Reason: Connection closed by the other side. - Connection closed by the other side.
logging-service-2-1 | [DefaultLogger] WARN at Connection: Connection{alive=false, connectionId=2, remoteAddress=hazelcast-2:5701} closed. Reason: Connection closed by the other side. - Connection closed by the other side.
logging-service-2-1 | [DefaultLogger] INFO at ConnectionManager: Removed connection to endpoint: hazelcast-2:5701:43fbc045-218c-4c9b-8a6a-eec75bb1a08a, connection: Connection{alive=false, connectionId=2, remoteAddress=hazelcast-2:5701}
logging-service-3-1 | [DefaultLogger] INFO at ConnectionManager: Removed connection to endpoint: hazelcast-2:5701:43fbc045-218c-4c9b-8a6a-eec75bb1a08a, connection: Connection{alive=false, connectionId=1, remoteAddress=hazelcast-2:5701}
logging-service-1-1 | [DefaultLogger] WARN at Connection: Connection{alive=false, connectionId=0, remoteAddress=hazelcast-2:5701} closed. Reason: Connection closed by the other side. - Connection closed by the other side.
logging-service-1-1 | [DefaultLogger] INFO at ConnectionManager: Removed connection to endpoint: hazelcast-2:5701:43fbc045-218c-4c9b-8a6a-eec75bb1a08a, connection: Connection{alive=false, connectionId=0, remoteAddress=hazelcast-2:5701}
```

Приклад POST запиту (інші аналогічні):

POST http://localhost:3000/transaction

Params Authorization Headers (10) Body Pre-request Script Tests Settings Cookies Beautify

none form-data x-www-form-urlencoded raw binary JSON

```
1 {
2   "user_Id": "test-3-1",
3   "amount": 10
4 }
```

Body Cookies Headers (5) Test Results Status: 200 OK Time: 76 ms Size: 241 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "transaction_ID": "b1f66378-0ad6-4894-9ee6-499eedfa9f6e",
3   "balance": 10
4 }
```

Логу logging-service:

```
logging-service-3-1 | {"level":30,"time":1774212815127,"pid":18,"hostname":"0f5d7c0a0330","instanceName":"logging-service-3","transaction_ID":"b1f66378-0ad6-4894-9ee6-499eedfa9f6e","user_Id":"test-3-1","amount":10,"msg":"Stored transaction in Hazelcast"}
logging-service-3-1 | {"level":30,"time":1774212839297,"pid":18,"hostname":"0f5d7c0a0330","instanceName":"logging-service-3","transaction_ID":"3369ed55-b827-43d2-b1d9-14512d069589","user_Id":"test-3-2","amount":10,"msg":"Stored transaction in Hazelcast"}
logging-service-2-1 | {"level":30,"time":1774212842505,"pid":18,"hostname":"2d46c924d670","instanceName":"logging-service-2","transaction_ID":"58ddc6cd-ae67-4382-8f49-c0710f0da25a","user_Id":"test-3-3","amount":10,"msg":"Stored transaction in Hazelcast"}
logging-service-1-1 | {"level":30,"time":1774212845656,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"51b424aa-c799-4ce3-ba2e-3fda5b32018f","user_Id":"test-3-4","amount":10,"msg":"Stored transaction in Hazelcast"}
logging-service-1-1 | {"level":30,"time":1774212849575,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"55bd7e37-e953-4436-9b09-2a29282ccb8c","user_Id":"test-3-5","amount":10,"msg":"Stored transaction in Hazelcast"}
```

GET /accounts:

GET http://localhost:3000/accounts

Params Authorization Headers (10) Body Pre-request Script Tests Settings Cookies

Body Cookies Headers (5) Test Results Status: 200 OK Time: 27 ms Size: 396 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "accounts": {
3     "test-1-1": 10,
4     "test-1-2": 10,
5     "test-1-3": 10,
6     "test-1-4": 10,
7     "test-1-5": 10,
8     "test-2-1": 10,
9     "test-2-2": 10,
10    "test-2-3": 10,
11    "test-2-4": 10,
12    "test-2-5": 10,
13    "test-3-1": 10,
14    "test-3-2": 10,
15    "test-3-3": 10,
16    "test-3-4": 10,
17    "test-3-5": 10
18  }
19 }
```

Приклад GET /user/<user_Id> (інші ідентичні):

GET http://localhost:3000/user/test-3-1

Status: 200 OK Time: 23 ms Size: 333 B

```
{  "balance": 10,  "logs": [    {      "transaction_ID": "b1f66378-0ad6-4894-9ee6-499eedfa9f6e",      "timestamp": "2026-03-22T20:53:35.090Z",      "msg": {        "user_Id": "test-3-1",        "amount": 10      }    }  ]}
```

- Вимкнення assignment_3-hazelcast-2 та assignment_3-hazelcast-3:

```
logging-service-3-1 | [DefaultLogger] WARN at Connection: Connection{alive=false, connectionId=1, remoteAddress=hazelcast-2:5701} closed. Reason: Connection closed by the other side. - Connection closed by the other side.
logging-service-2-1 | [DefaultLogger] WARN at Connection: Connection{alive=false, connectionId=2, remoteAddress=hazelcast-2:5701} closed. Reason: Connection closed by the other side. - Connection closed by the other side.
logging-service-2-1 | [DefaultLogger] INFO at ConnectionManager: Removed connection to endpoint: hazelcast-2:5701:43fbc045-218c-4c9b-8a6a-ec75bb1a08a, connection: Connection{alive=false, connectionId=2, remoteAddress=hazelcast-2:5701}
logging-service-3-1 | [DefaultLogger] INFO at ConnectionManager: Removed connection to endpoint: hazelcast-2:5701:43fbc045-218c-4c9b-8a6a-ec75bb1a08a, connection: Connection{alive=false, connectionId=1, remoteAddress=hazelcast-2:5701}
logging-service-1-1 | [DefaultLogger] WARN at Connection: Connection{alive=false, connectionId=0, remoteAddress=hazelcast-2:5701} closed. Reason: Connection closed by the other side. - Connection closed by the other side.
logging-service-1-1 | [DefaultLogger] INFO at ConnectionManager: Removed connection to endpoint: hazelcast-2:5701:43fbc045-218c-4c9b-8a6a-ec75bb1a08a, connection: Connection{alive=false, connectionId=0, remoteAddress=hazelcast-2:5701}
```

```
logging-service-3-1 | [DefaultLogger] WARN at Connection: Connection{alive=false, connectionId=0, remoteAddress=hazelcast-3:5701} closed. Reason: Connection closed by the other side. - Connection closed by the other side.
logging-service-3-1 | [DefaultLogger] INFO at ConnectionManager: Removed connection to endpoint: hazelcast-3:5701:186eb59b-e65d-446a-be37-83b4aa77e48c, connection: Connection{alive=false, connectionId=0, remoteAddress=hazelcast-3:5701}
logging-service-1-1 | [DefaultLogger] WARN at Connection: Connection{alive=false, connectionId=1, remoteAddress=hazelcast-3:5701} closed. Reason: Connection closed by the other side. - Connection closed by the other side.
logging-service-2-1 | [DefaultLogger] WARN at Connection: Connection{alive=false, connectionId=0, remoteAddress=hazelcast-3:5701} closed. Reason: Connection closed by the other side. - Connection closed by the other side.
logging-service-1-1 | [DefaultLogger] INFO at ConnectionManager: Removed connection to endpoint: hazelcast-3:5701:186eb59b-e65d-446a-be37-83b4aa77e48c, connection: Connection{alive=false, connectionId=1, remoteAddress=hazelcast-3:5701}
logging-service-2-1 | [DefaultLogger] INFO at ConnectionManager: Removed connection to endpoint: hazelcast-3:5701:186eb59b-e65d-446a-be37-83b4aa77e48c, connection: Connection{alive=false, connectionId=0, remoteAddress=hazelcast-3:5701}
```

Приклад POST запиту (інші аналогічні):

POST http://localhost:3000/transaction

Status: 200 OK Time: 29 ms Size: 241 B

```
{  "transaction_ID": "1f82b800-b8f2-4b6e-b2c4-4b026583fce3",  "balance": 10}
```


Λογυ logging-service:

```
logging-service-2-1 | {"level":30,"time":1774213294027,"pid":18,"hostname":"2d46c924d670","instanceName":"logging-service-2","transaction_ID":"1f82b800-b8f2-4b6e-b2c4-4b026583fce3","user_Id":"test-4-1","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-1-1 | {"level":30,"time":1774213545732,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"07df029e-c978-4bad-82da-9746b3ead283","user_Id":"test-4-2","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-1-1 | {"level":30,"time":1774213549153,"pid":18,"hostname":"8c1d6018e00e","instanceName":"logging-service-1","transaction_ID":"47fb45da-96be-4c6d-89c5-eb9aaefa9f5","user_Id":"test-4-3","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-3-1 | {"level":30,"time":1774213551759,"pid":18,"hostname":"0f5d7c0a0330","instanceName":"logging-service-3","transaction_ID":"2ba61e1e-566c-4fca-9bd6-125042e5d059","user_Id":"test-4-4","amount":10,"msg":"Stored transaction in Hazelcast"}
```

```
logging-service-2-1 | {"level":30,"time":177421355259,"pid":18,"hostname":"2d46c924d670","instanceName":"logging-service-2","transaction_ID":"84a563ea-fb07-465b-95fb-04f8fd0e0bf4","user_Id":"test-4-5","amount":10,"msg":"Stored transaction in Hazelcast"}
```

GET /accounts:

GET http://localhost:3000/accounts

Status: 200 OK Time: 55 ms Size: 466 B

Body

```
{
  "accounts": {
    "test-1-1": 10,
    "test-1-2": 10,
    "test-1-3": 10,
    "test-1-4": 10,
    "test-1-5": 10,
    "test-2-1": 10,
    "test-2-2": 10,
    "test-2-3": 10,
    "test-2-4": 10,
    "test-2-5": 10,
    "test-3-1": 10,
    "test-3-2": 10,
    "test-3-3": 10,
    "test-3-4": 10,
    "test-3-5": 10,
    "test-4-1": 10,
    "test-4-2": 10,
    "test-4-3": 10,
    "test-4-4": 10,
    "test-4-5": 10
  }
}
```

Παράδειγμα GET /user/<user_Id> (ή άλλοι ιδιότητες):

GET http://localhost:3000/user/test-4-1

Status: 200 OK Time: 47 ms Size: 333 B

Body

```
{
  "balance": 10,
  "logs": [
    {
      "transaction_ID": "1f82b800-b8f2-4b6e-b2c4-4b026583fce3",
      "timestamp": "2026-03-22T21:01:34.022Z",
      "msg": {
        "user_Id": "test-4-1",
        "amount": 10
      }
    }
  ]
}
```

3. Тестування продуктивності:

Тестування відбувалося за допомогою модуля load-test-client.

- Сценарій 1: 10 Користувачів, 10000 запитів кожен:

```
-----
RESULTS:
-----
Total requests: 100000
Successful requests: 100000
Failed requests: 0
Total time: 421.75 seconds
Requests per second: 237.11 req/s

Service timing breakdown:
  Logging service: 951.87 s (total time for all calls, 100000 calls, avg: 9.52 ms)
  Counter service: 3402.68 s (total time for all calls, 100001 calls, avg: 34.03 ms)
  Logging contribution: 21.86%
  Counter contribution: 78.14%

Final balances:
  user-1: 10000
  user-10: 10000
  user-2: 10000
  user-3: 10000
  user-4: 10000
  user-5: 10000
  user-6: 10000
  user-7: 10000
  user-8: 10000
  user-9: 10000
-----
```

- Сценарій 2: 10 Користувачів, 10000 запитів на спільний рахунок:

```
-----
RESULTS:
-----
Total requests: 100000
Successful requests: 100000
Failed requests: 0
Total time: 716.09 seconds
Requests per second: 139.65 req/s

Service timing breakdown:
  Logging service: 267.89 s (total time for all calls, 100000 calls, avg: 2.68 ms)
  Counter service: 6865.23 s (total time for all calls, 100001 calls, avg: 68.65 ms)
  Logging contribution: 3.76%
  Counter contribution: 96.24%

Final balances:
  shared-account: 100000
-----
```

Спостерігаємо суттєво довший час роботи в обох випадках порівняно з **Task 1**, де обидва сценарії зайняли приблизно **120** секунд. Це можна пояснити додатковою роботою, затраченою на комунікацію з Hazelcast та з базою даних. Крім цього спостерігаємо:

- Порівняно з **Task 1**, де **logging** та **counter** займали приблизно однакову кількість часу, зараз у обох випадках **counter** займає суттєво більше часу - схоже, робота з **PostgreSQL** все ж обчислювально важча, ніж з **Hazelcast**.
- В **Task 1** ми спостерігали майже однакову продуктивність для обох сценаріїв - зараз же картини стали дуже відрізнятися:
 - По-перше, хоч **counter** в обох випадках є основним блокером, в сценарії 2 він займає вдвічі більше часу - не дивно, адже там всі клієнти звертаються до одного і того ж користувача, відповідно до одного і того ж запису в базі даних, що збільшує навантаження на неї і не дає опрацьовувати незалежні виклики одночасно.
 - По-друге, цікава динаміка **logging** сервісу - у сценарії 1 він займає майже вдвічі більше, ніж у **Task 1**, а в сценарії 2 - навпаки, майже вдвічі менше. Припускаю, на це можуть впливати внутрішні оптимізації Hazelcast для опрацювання послідовних викликів до одних і тих же записів у хеш-таблиці, які допомагають компенсувати overhead від Hazelcast у другому сценарії.